
Emergent Ordinal Geometry in Transformers Trained on Local Comparisons

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Transitive inference is the challenge of inferring that $A < C$ from knowing only
2 adjacent relations ($A < B$, $B < C$). It is solved by humans and animals not through
3 logical chaining but via an analogue mental number line, whose signature is the
4 symbolic distance effect: distant comparisons are easier than nearby ones. We ask
5 whether Transformers acquire the same primitive, training small models exclusively
6 on adjacent comparisons from a hidden total order and evaluating generalization
7 to unseen distant pairs. We find that out-of-distribution generalization emerges
8 alongside a striking geometric reorganization: entity embeddings collapse onto
9 a one-dimensional manifold whose principal axis recovers the hidden rank order
10 with near-perfect fidelity, and this structure is sensitive to optimization in ways
11 that produce grokking-like transient dynamics. Critically, even when accuracy is
12 at ceiling, decision confidence and geometric separation both scale monotonically
13 with rank distance, directly mirroring the symbolic distance effect observed across
14 decades of behavioural experiments on humans, primates, and rodents. These
15 results ground a 50-year-old behavioural regularity in the geometry of learned
16 representations, offering a mechanistic account of transitive inference that bridges
17 cognitive science and modern neural networks.

18 1 Introduction

19 Knowing that Alice is taller than Bob and Bob taller than Carol, one infers that Alice is taller
20 than Carol without direct comparison. This capacity, *transitive inference* (TI), is phylogenetically
21 widespread and documented in children, primates, rodents, and birds (Bryant and Trabasso [1971],
22 Steirn et al. [1995]) and, crucially, is not solved by formal logic. Behavioral work instead supports
23 an analog mental number line: items are arranged along a continuous internal ordering from which
24 comparisons are read off spatially (Riley [1976]). The signature of this account is the symbolic
25 distance effect: distant comparisons are faster and more accurate than adjacent ones (Bryant and
26 Trabasso [1971]), the opposite of what step-by-step chaining predicts, and exactly what one expects
27 if order is encoded as position on a line.

28 Transformers are under no explicit pressure to represent order this way, yet abstract relational structure
29 is known to emerge in their representations as a byproduct of training (Minegishi et al. [2026]),
30 sometimes abruptly and late, in grokking-like transitions (Power et al. [2022]). We ask a question at
31 the intersection of cognitive science and interpretability: when a Transformer learns to order items
32 from only local comparisons, does it discover the same primitive, *a mental number line*, and exhibit
33 the same behavioral signatures as biological cognition?

34 Mirroring the original Bryant and Trabasso [1971] paradigm, in which subjects learn adjacent
35 comparisons and are tested on inferred distant pairs, we train small Transformers only on adjacent
36 relations from a hidden total order and evaluate on held-out distant pairs. We find that out-of-
37 distribution generalization coincides with a geometric reorganization: entity embeddings collapse onto

a one-dimensional manifold whose principal axis recovers the hidden rank order. The resemblance is behavioral as well as structural, even at accuracy ceiling, decision confidence and geometric separation scale with rank distance, a computational analogue of the symbolic distance effect. Our contributions are: **(i)** A synthetic task, modeled on the classic TI paradigm, isolating ordinal generalization from local comparisons. **(ii)** Evidence that Transformers spontaneously form a 1D ordinal manifold and reproduce the symbolic distance effect.

2 Background

Transitive inference and the distance effect: Since Bryant and Trabasso [1971], TI has been studied as ordinal rather than logical reasoning; the symbolic distance and end-anchor effects are its canonical behavioral signatures across species Steirn et al. [1995].

Emergent structure in Transformers: Generalizing solutions can appear abruptly after memorization, a phenomenon termed *grokking* (Power et al. [2022]), driven by weight decay and trackable through internal progress measures as talked about in texts like Nanda et al. [2023]; Minegishi et al. [2026] formalize relational reasoning as emergent representational geometry.

Linear representations: Semantic and relational structure is often encoded linearly in embedding space Park et al. [2025], motivating our use of principal-component analysis as a probe.

3 Task and Methods

Setup: Let $E = \{e_0, \dots, e_{N-1}\}$ carry a hidden total order $e_0 \prec e_1 \prec \dots \prec e_{N-1}$. Each entity is an atomic token and the order is never given explicitly. A comparison is a triple $(e_i, ?, e_j)$ whose target is the relation

$$r(e_i, e_j) = \begin{cases} \prec; & i < j, \\ \succ; & i > j. \end{cases} \quad (1)$$

Train/OOD split: The model is trained *only* on adjacent comparisons, in both directions,

$$\mathcal{D}_{\text{train}} = \{(e_i, e_{i+1}), (e_{i+1}, e_i) : 0 \leq i < N - 1\}, \quad (2)$$

and evaluated on the disjoint held-out set of all distant pairs,

$$\mathcal{D}_{\text{OOD}} = \{(e_i, e_j) : |i - j| > 1\}, \quad \mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{OOD}} = \emptyset \quad (3)$$

Since no distant pair is ever observed, success on $\mathcal{D}_{\text{OOD}}$ requires reconstructing the global order $\prec$ from local comparisons alone, the computational form of the Bryant–Trabasso paradigm.

Model: We use a single-layer Transformer f_θ with entity embeddings $\{h_e \in \mathbb{R}^d\}$, one attention head, and unembedding head W_U . The relation is read from the final-position residual; we train with cross-entropy on the relation token using AdamW with weight decay λ .

Measuring the number line. Let $H = [h_{e_0}; \dots; h_{e_{N-1}}] \in \mathbb{R}^{N \times d}$ be the entity embeddings and u_1 their first principal component, with eigenvalues $\lambda_1 \geq \dots \geq \lambda_d$. We quantify ordinal structure three ways:

1. **Linearity:** the variance fraction on u_1 , $\rho_1 = \lambda_1 / \sum_k \lambda_k$, large when embeddings lie near a line.

2. **Rank alignment:** the correlation between projected position $\pi_i = \langle h_{e_i}, u_1 \rangle$ and true rank i ,

$$r_{\text{rank}} = \text{corr}(\{\pi_i\}_i, \{i\}_i), \quad (4)$$

near ± 1 when the line recovers the hidden order.

3. **Geometric separation:** $g_{ij} = |\pi_i - \pi_j|$, the distance between two items on the recovered line.

74 **Behavioral signatures.** Because accuracy saturates once the order is learned, we measure graded
75 effects that persist at ceiling. For a query (e_i, e_j) with correct and incorrect relation tokens c, w , the
76 *decision confidence* is the logit margin

$$m_{ij} = (W_U z_{ij})_c - (W_U z_{ij})_w, \quad (5)$$

77 where z_{ij} is the final-position residual. The symbolic distance effect predicts that both m_{ij} and g_{ij}
78 increase with rank distance $|i - j|$; we report their Pearson correlation with $|i - j|$ over $\mathcal{D}_{\text{OOD}}$.

79 4 Results

80 The synthetic model is a single-layer, single-block Transformer with hidden dimension $d = 32^1$,
81 2 attention heads, a $4d$ -wide GELU MLP, learned positional embeddings, and pre-LayerNorm;
82 the relation is decoded from the final-position residual by a linear unembedding head ($\approx 5 \times 10^4$
83 parameters). We optimize the relation-token cross-entropy with AdamW (learning rate 10^{-3} , batch
84 size 64) and weight decay λ swept per chain length N , training for up to 5×10^4 steps and reporting
85 all quantities at the peak-OOD accuracy checkpoint.

86 We run the experiment for $N = 10, 15$, and 20 entities. The figures for $N = 10, 20$ can be found in the
87 appendix. The following discussion is a qualitative breakdown of the trends seen in all three results.

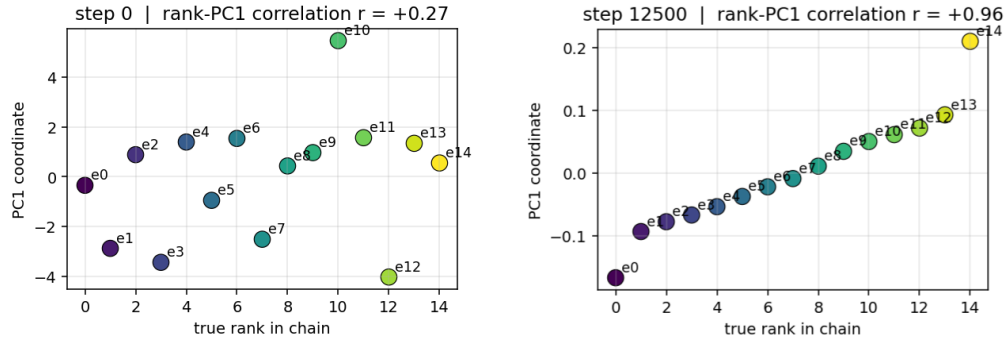


Figure 1: First principle projection vs. true rank in chain for $N = 15$ entities.

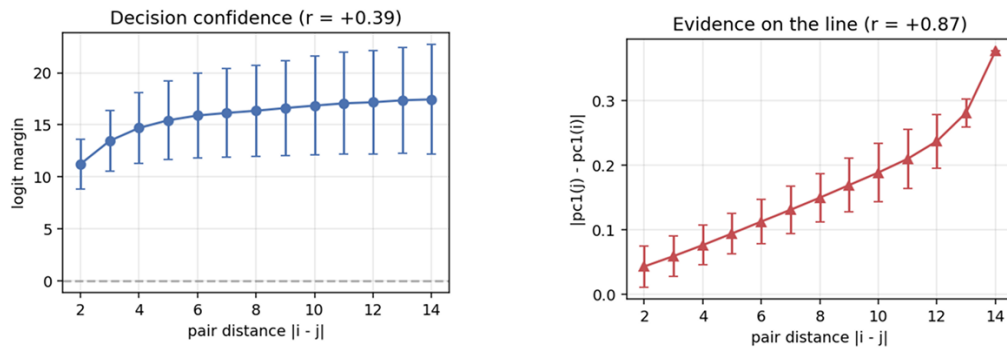


Figure 2: (a) Decision confidence as a function of distance between compared entities. (b) PC1 distance as a function of pair distance.

88 Figure 1 (rank vs. PC1) makes the linearity argument explicit by plotting each entity’s projection
89 onto the first principal component against its true rank: at initialization the relationship is essentially
90 random, whereas at peak the points form a clean monotonic staircase, confirming that the principal

¹We use $d = 64$ for reliable grokking when training with more entities.

axis recovers the hidden total order almost exactly. Figure 2 tests for the symbolic distance effect using measures that remain informative once accuracy saturates: decision confidence and geometric separation on the recovered line, both increase monotonically with rank distance, exactly as predicted if comparisons are read off a mental number line.

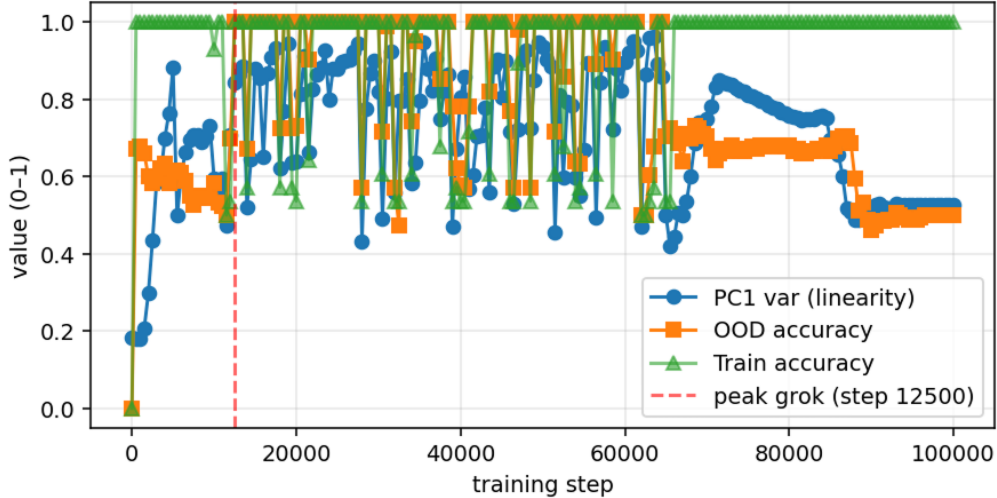


Figure 3: Training dynamics for $N = 15$ entities.

Figure 3 overlays PC1 variance, OOD accuracy, and train accuracy across training; train accuracy saturates almost immediately while OOD generalization and embedding linearity rise together and later, and crucially both degrade after the peak (dashed line), illustrating that the structured solution is transient under fixed weight decay — which is why all analyses are reported at the peak-OOD checkpoint.

5 Limitations

Our study has a few limitations worth noting. Our evidence is correlational rather than causal: we show that the leading principal component recovers the rank order and that confidence and geometric separation scale with rank distance, but we leave to future work the interventions needed to confirm the model relies on this axis. Our results also come from small single-layer Transformers on a synthetic linear-order task, so questions of robustness across seeds and architectures, and whether the same geometry arises for non-linear or cyclic relations, remain open avenues to explore.

6 Conclusion and Future Work

This short paper is a study of the existence of a cognitive primitive present in conscious, living organisms in a transformer. The empirical proofs presented in the paper point to the existence of a transitive inference algorithm common between organisms and transformers. Transformers represent relational information on a straight line, and are more confident in the relationships of entities far along in the chain than close together. The generalisation behaviour is transient, consistent with findings in the vast literature of grokking.

The study of this phenomenon is far from complete, and there are a lot of rich directions to explore from this result. This does the same representation arise in machine learning architectures other than transformers? The behavior of LLMs is also of interest, since they have richer architectures and are fed richer datasets during training. For relationships which are not linear by nature but still display some relational interplay² does a transformer internalise equivalent representations? And finally, this study does not make any claims about the causality of this representation.

²Months of the year, for example, are cyclic - January is both before and after December.

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141 A Technical Appendices and Supplementary Material

142 A.1 Data for $N = 10$ entities

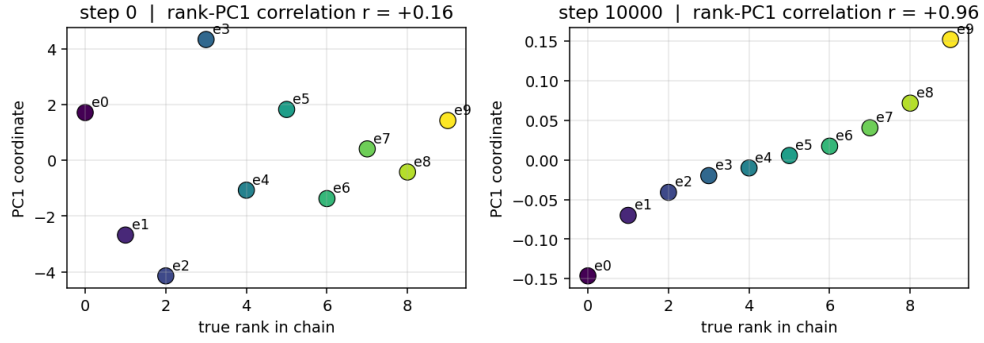


Figure 4: First principle projection vs. true rank in chain for $N = 10$ entities.

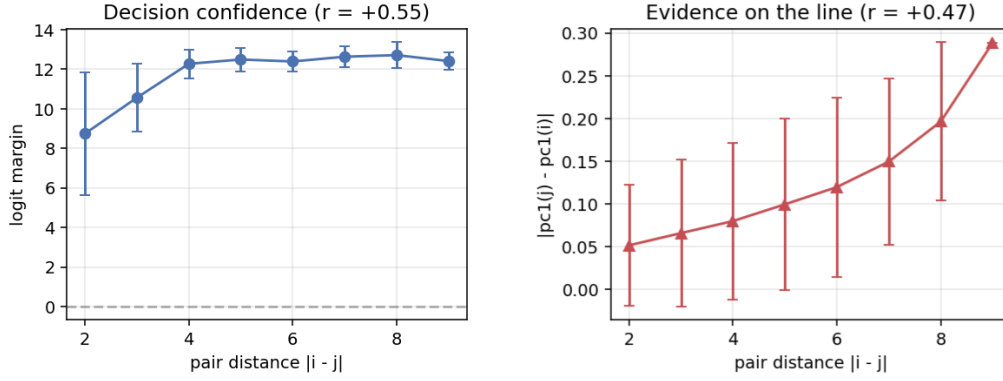


Figure 5: (a) Decision confidence as a function of distance between compared entities. (b) PC1 distance as a function of pair distance.

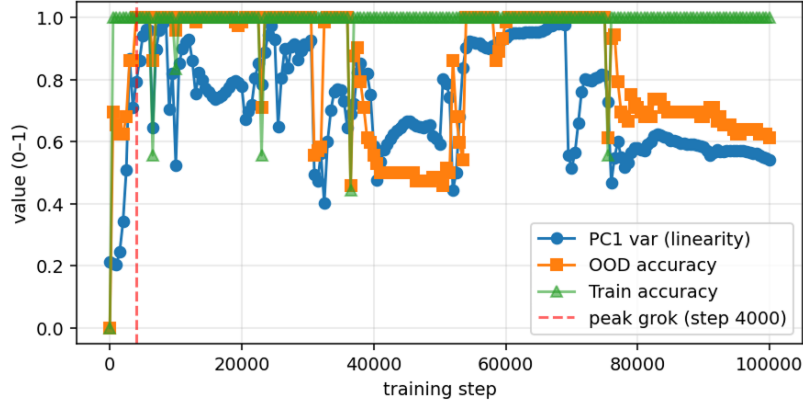


Figure 6: Training dynamics for $N = 10$ entities.

143 A.2 Data for $N = 20$ entities

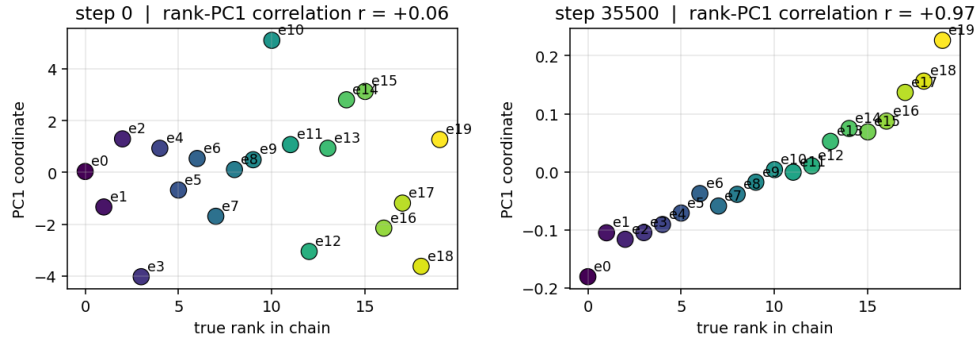


Figure 7: First principle projection vs. true rank in chain for $N = 10$ entities.

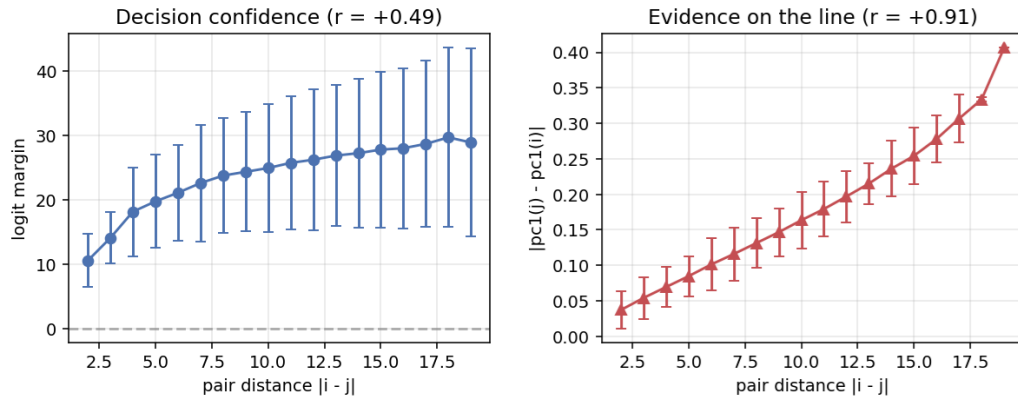


Figure 8: (a) Decision confidence as a function of distance between compared entities. (b) PC1 distance as a function of pair distance.

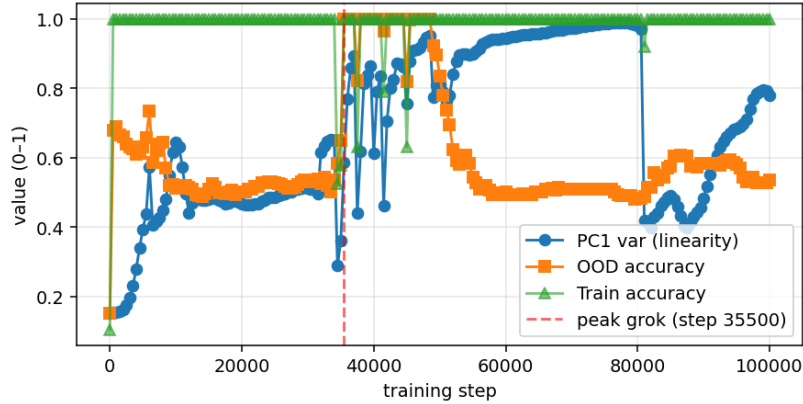


Figure 9: Training dynamics for $N = 10$ entities.

Conference For AI Scientists 2026 - AI Involvement Checklist

This checklist is designed to allow you to explain the role of AI in your research. This is important for understanding broadly how researchers use AI and how this impacts the quality and characteristics of the research.

The checklist has two parts: **Research Stage Assessment** (items 1-4), where you rate the level of AI involvement and iteration effort at each stage of the research process, and **AI System Documentation** (items 5-6), where you provide free-text descriptions of the AI systems used and their observed limitations.

IMPORTANT, please:

- **Do not remove the checklist! Papers not including the checklist will be desk rejected.**
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- **Keep all checklist questions/answers and guidelines.**
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For items 1-4, give a score from the scale below that defines the role of AI in each part of the scientific process. The scores are as follows:

- **[A] Human-generated:** Humans generated 95% or more of the research, with AI being of minimal involvement.
- **[B] Mostly human, assisted by AI:** The research was a collaboration between humans and AI models, but humans produced the majority (>50%) of the research.
- **[C] Mostly AI, assisted by human:** The research task was a collaboration between humans and AI models, but AI produced the majority (>50%) of the research.
- **[D] AI-generated:** AI performed over 95% of the research. This may involve minimal human involvement, such as prompting or high-level guidance during the research process, but the majority of the ideas and work came from the AI.

For each research stage where AI was involved, i.e. where you selected **[B]**, **[C]**, or **[D]**, please also indicate the approximate level of iteration effort required using the provided iteration macros. Count substantive attempts, prompts, runs, agent trajectories, or course corrections; do not count minor wording edits.

- **[Low]** : approximately 1-10 substantive attempts.

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- **[Unclear]** : the authors cannot reasonably estimate.

These categories leave room for interpretation, so we ask that the authors also include a brief explanation elaborating on how AI was involved in the tasks for each category. Please keep your explanation to less than 150 words.

1. **Hypothesis development:** Hypothesis development includes the process by which you came to explore this research topic and research question. This can involve the background research performed by either researchers or by AI. This can also involve whether the idea was proposed by researchers or by AI.

Answer: **[A]**

Iteration Effort: **[NA]**

Explanation: The original problem statement was inspired by new research in the field, AI was utilised to clear up structural inconsistencies in the problem statement after being formed.

2. **Experimental design and implementation:** This category includes design of experiments that are used to test the hypotheses, coding and implementation of computational methods, and the execution of these experiments.

Answer: **[B]**

Iteration Effort: **[Medium]**

Explanation: Experiment design was thought of by humans, but the granular design choices were made through iterating with AI.

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Answer: **[D]**

Iteration Effort: **[Medium]**

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For items 5-6, provide a free-text description.

5. **AI systems used:** Describe all AI systems used in this research without naming specific products or models. Use system-level descriptors only (e.g., “a frontier large language model”, “a protein structure prediction system”, “a custom multi-agent framework built on open-source models”). Include relevant details such as system type, scale, and capabilities. **Specific model and product names should only be included in the camera-ready version.**

Description: Frontier large language model

6. **Observed AI Limitations:** What specific limitations and failure modes have you observed when using AI as a partner or lead author?

Description: While granular reasoning is almost always sound, high level reasoning, such as why the experiment works/doesn't work is almost always framed in the wrong light.

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The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

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The checklist answers are an integral part of your paper submission. They are visible to the reviewers and area chairs. You will be asked to also include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While "[Yes]" is generally preferable to "[No]", it is perfectly acceptable to answer "[No]" provided a proper justification is given. In general, answering "[No]" or "[NA]" is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

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